Ontology-Based Conditional Generative Adversarial Networks for the Generation of Personal Health Data

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***Abstract*— This paper presents Onto-CGANS and OntoZSL-CGANS, a novel approach for generating realistic personal health data by incorporating ontology embedding and zero-shot learning techniques. Personal health data generation is crucial for scientific research, but privacy concerns and limited accessibility hinder its availability. Existing data-driven methods struggle to generate high-quality data for unseen classes. Onto-CGANS and OntoZSL-CGANS address these challenges by leveraging ontological schema embedding and zero-shot learning to generate sparse and unseen patient data, surpassing the limitations of current generative methods. Experimental results demonstrate the effectiveness of Onto-CGANS in generating realistic data, while ZSL is still in its early stages of research and requires further refinement and application of the proposed framework.**

***Index Terms*—DP-CGAN, Ontologies, Knowledge Graphs, Zero-Shot Learning, GANs, Personal Health Data, Large Language Models**

I. Introduction

Personal health data has proven to be of great value for scientific research purposes [1]. However, the sensitivity and limited accessibility of this data often pose challenges for researchers. Even when data is accessible through request, gaining access to it requires a significant investment of time and effort, without a guarantee that the data will be suitable for the intended research purposes [2].

To address this challenge, one promising approach is the generation of synthetic personal health data. A recent study introduced DP-CGANs, a Generative Adversarial Network (GAN) framework that combines differential privacy methods to generate privacy-preserving and realistic personal health data [3]. However, this fully data-driven approach faces difficulties in generating high-quality synthetic data when real data is scarce or unavailable. Generating patient data for rare diseases, in particular, remains a significant challenge due to the sparsity of such data in reality [4], [5].

Over the past decades, researchers have made significant advancements in understanding human health and diseases. This has led to the idea of combining human knowledge with data-driven methods, such as GAN models, to enhance the generation of personal health data [6]–[9].

This paper proposes an extension to the DP-CGANs framework called Onto(ZSL)-CGANs, which incorporates semantic embedding techniques for ontological schema. By leveraging ontologies, Onto(ZSL)-CGANs enable the generation of sparse and unseen patient data for rare diseases, addressing the limitations of existing generative methods.

To investigate the capabilities and performance of Onto(ZSL)-CGANs, this research aims to address the following research questions:

1) How can related ontologies be embedded into a GAN model to generate realistic personal health data?
2) How does an ontology-based GAN perform compared to a baseline GAN in generating personal health data?
3) How could zero-shot learning (ZSL) be used for the classification of generated synthetic personal health data using a GAN model on unseen classes?

The structure of this paper is as follows: Section II provides a literature review on GANs, ontologies, and ZSL, along with an overview of state-of-the-art methods. Section III describes the proposed methodology and its practical implementation. Section IV presents the experimental setup and results, followed by a discussion of the findings in section VI. Finally, section VII concludes the paper, highlighting the contributions of this work and suggesting possibilities for future research.

II. Related Work

A. Prior Knowledge for DP-CGANS

Differential Private Conditional Generative Adversarial Networks (DP-CGANs) consist of two neural network models: a generative model called the generator, and a discriminative model referred to as the discriminator [3]. The discriminator's objective is to distinguish between real and synthetic data, while the generator aims to produce synthetic data that is indistinguishable from real data sources [3]. These models engage in a competitive process where the discriminator is trained to maximize the log-likelihood of accurately classifying real and synthetic data, while the generator is trained to minimize the log-likelihood of the discriminator's correct classification [10].

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When dealing with tabular data in GANs, the dataset often contains both continuous and categorical variables, requiring distinct transformation techniques. Categorical variables are typically encoded using one-hot encoding, while continuous variables undergo mode-specific normalization [3], [11].

To handle imbalanced data, a conditional vector is introduced during the sampling process, utilizing the training-by-sampling method [3], [11].

The use of GANs for training on personal or sensitive data raises privacy concerns [12], [13]. To address these concerns, a differential privacy approach is employed to safeguard the privacy of both the discriminator and the generator. This involves augmenting the penalty gradient of the Wasserstein distance with Gaussian noise during the training of the discriminator [3], [14]. Additionally, the Rényi Differential Privacy (RDP) Account is utilized to derive more precise estimations of the privacy budget [3], [15].

B. Ontology Embeddings and Knowledge Graphs

In recent years, ontology embeddings have garnered significant attention, particularly in the medical and biological fields, as a means to represent entities and their relationships within vector spaces [16]–[18].

The process of embedding ontologies can be achieved through various methods, including projection, random walks, and knowledge graph embeddings [17]. Hybrid solutions combining these methods have also been explored. One popular approach is to combine projection and random walks, employing a two-step approach to ontology embedding. This enables leveraging the complementary strengths of these methods and capturing a more comprehensive representation of the ontology [17].

1) Ontology Projection: The initial step in ontology embedding is ontology projection, which transforms the ontology into a graph structure, preserving important adjacency information and relationships between entities and concepts. This captures the structural and semantic aspects of the ontology, enabling the utilization of graph-based algorithms and techniques for subsequent analysis. Ontology projection involves considering axioms of the form $A \subset B$, where A and B represent ontology classes. This generates the triple $\langle A, \text{subClassOf}, B \rangle$ [17]. Additionally, relations can be taken into account, allowing for the consideration of $A \subset B$ as well as $A \subset \exists R.B$, resulting in the triple $\langle A, R, B \rangle$ [17].

a) Taxonomies: Ontology projection achieves the generation of triples by considering axioms of the form $A \subset B$, where A and B represent ontology classes. This generates the triple $\langle A, \text{subClassOf}, B \rangle$ [17]. Additionally, relations can be taken into account, allowing for the consideration of $A \subset B$ as well as $A \subset \exists R.B$, resulting in the triple $\langle A, R, B \rangle$ [17].

b) DL2VEC: For ontology projection, different types of description logic, specifically TBOX axioms, are converted into a graph representation using a set of rules, as shown in Table I [19]. In some cases, the information may be presented as ABOX axioms of the form $C(a), \exists R, C(a), R(a, b)$, which provide additional rules as illustrated in Table II [19].

c) OWL2Vec:* Ontology projection takes into consideration the graph structure, lexical information, and logical constructors by employing a set of rules outlined in Table III [18].

2) Random Walk Embedding: On the other hand, random walks play a crucial role in capturing the contextual information and relationships within the ontology. By traversing the graph representation through random walks, we can explore the local and global neighborhoods of concepts and relations, capturing their semantic context and interconnections. This exploration allows us to generate meaningful vector representations that reflect the semantic relationships and similarities between entities [17], [20], [21].

Various approaches can be adopted for conducting random walks, including:

a) DeepWalk: DeepWalk combines language model techniques with deep learning to acquire implicit representations by treating walks as equivalent to sentences. It focuses on unbiased walks, considering only the local neighborhoods of the nodes [20]. A standard skip-gram model is utilized on the random walks, treating each node equally [20].

b) Node2Vec: Node2Vec employs techniques to obtain continuous feature representations for nodes by mapping them to a lower-dimensional feature space that maximizes the likelihood of preserving the network structure. It utilizes biased walks, taking into account both local neighborhoods and the global structure of the graph [21].

A modified skip-gram model is employed based on the random walks, considering only the nodes with the biased random walk. Weights are applied to nodes based on the probability of their visibility during random walks, enabling the capture of both local neighborhoods and the global graph structure [21].

3) Knowledge Graph Embedding: In addition to random walks, Knowledge Graph Embedding (KGE) is another prominent technique employed for representing and capturing the semantics of concepts and relations in ontologies. KGE offers an alternative approach to random walks by encoding entities and relationships into low-dimensional continuous vector spaces, enabling efficient processing and analysis of graph data [17], [22].

Two common approaches for KGE are neural network-based methods and matrix factorization techniques [23].

Neural network-based methods leverage deep learning architectures to model the complex patterns and interactions within knowledge graphs. These methods often employ techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph convolutional networks (GCNs) to capture the structural and semantic information of entities and relationships. Some popular models in this category include ConvE and RotatE [22], [24].

Matrix factorization techniques, on the other hand, factorize the knowledge graph into low-rank matrices to obtain the embeddings. These methods typically utilize matrix factorization algorithms such as singular value decomposition (SVD) or non-negative matrix factorization (NMF) to decompose the adjacency matrix of the graph. Some popular models in this

Condition 1	Condition 2	Projected Triples
$A \subset QR_0 \dots R_m D$	$D := B_1 \vee \dots \vee B_n \mid B_1 \wedge \dots \wedge B_n$	$\langle A, (R_0 \dots R_m), B_i \rangle$ for $i \in 1 \dots n$
$A \equiv QR_0 \dots R_m D$	$D := B_1 \vee \dots \vee B_n \mid B_1 \wedge \dots \wedge B_n$	$\langle A, (R_0 \dots R_m), B_i \rangle$ for $i \in 1 \dots n$
$A \subset B$		$\langle A, SubClassOf, B \rangle$
$A \equiv B$		$\langle A, EquivalentTo, B \rangle$

TABLE I: The parsing rules DL2Vec uses for TBox Axioms. [19] [17]

Condition	Projected Triples
Conditions from table I	Projected Triples from table I
$C(a)$	$\langle a$, http://type, $C \rangle$
$\exists R.C(a)$	$\langle a, R, C \rangle$
$R(a, b)$	$\langle a, R, b \rangle$

TABLE II: The extended parsing rules DL2Vec uses for ABox Axioms. [17]

Condition 1	Condition 2	Projected Triples
$A \subset \Box r.D$	$D \equiv B \mid B_1 \vee \dots \vee B_n \mid B_1 \wedge \dots \wedge B_n$	$\langle A, r, B \rangle$
$\Box r.D \subset A$	$D \equiv B \mid B_1 \vee \dots \vee B_n \mid B_1 \wedge \dots \wedge B_n$	$\langle A, r, B \rangle$
$\exists r.T \subset A$ (domain)	$T \subset \forall r.B$ (range)	$\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$A \subset \exists R.\{b\}$	$B(b)$	$\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$r \subset r^{'}$	$\langle A, r^{'}, B \rangle$ has been projected	$\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$r^{'} \equiv r^{-}$	$\langle B, r^{'}, A \rangle$ has been projected	$\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$s_1 o \dots o s_n \subset r$	$\langle A, s_1, C_1 \rangle \dots \langle C_n, s_n, B \rangle$ have been projected	$\langle A, r, B_i \rangle$ for $i \in 1, \dots, n$
$B \subset A$	-	$\langle B, , A \rangle$ $\langle$, rdfs:subClassOf^{-}, $B \rangle$
$A(a)$	-	$\langle a$, rdf:type, $A \rangle$ $\langle A$, rdf:type^{-}, $a \rangle$
$r(a, b)$	-	$\langle a, r, b \rangle$

TABLE III: The parsing rules OWL2Vec* uses. [18] [17]

category include TransE, TransR, and ComplEx [22], [25]–[27].

One notable implementation of KGE is PyKEEN, a Python library that provides a comprehensive framework for KGE. PyKEEN offers a collection of state-of-the-art embedding models, evaluation metrics, and utilities to facilitate the process of KGE [22].

C. Zero-shot Learning

Numerous studies have demonstrated that the utilization of ontology embeddings, in conjunction with ZSL, yields comparable, if not improved performances compared to state-of-the-art methods [8], [16]. Additionally, this approach enables the generation of data for previously unseen classes [8], [9], [16], [28].

Unseen classes pose a challenge in datasets due to their infrequent occurrence, making it difficult to capture them solely through ontological embedding [8], [16], [29], [30]. This is where ZSL becomes crucial.

The key distinction between traditional supervised learning and unsupervised ZSL lies in the fact that ZSL must classify synthetic data on which it has not been trained [8], [16]. This is achieved by utilizing semantic information and attributes of seen classes, as well as leveraging shared attributes between seen and unseen classes [16].

Several techniques have been employed to facilitate ZSL implementations with GANs, including:

1) Standard GAN with no Zero-shot classifier: A straightforward approach involves using the basic structure of an ontological GAN, where ZSL is implicitly inferred from the behavior of the generative model. The generative model learns to generate new triples consistent with the knowledge graph from its embedded model, while the discriminator is trained to distinguish between real and synthetic triples [31]. By providing the generator with an unseen target class and allowing it to generate a new triple with that class as the subject, this method enables ZSL of new relationships between classes that were not originally present [31].

2) Auto-Encoder with a Zero-shot Classifier: A more sophisticated approach involves using an auto-encoder variant in combination with a zero-shot classifier [16]. The generator operates in a similar manner as described in section II-C1, while the discriminator is an extension of the typical implementation, as explained in section II-C1, but now incorporates classification loss and pivot regularization to ensure inter-class discrimination of the generated features [16].

Additionally, a zero-shot classifier is introduced, which, similar to the discriminator, takes samples from the generator but aims to classify the unseen classes for testing purposes, thereby enabling the generative model to generate data for unseen classes. The zero-shot classifier shares the same objective as the generator, minimizing the log-likelihood for the discriminator to accurately predict real and synthetic data, thereby continuously improving its performance [16].

3) Multi-Layer Perceptron with a Zero-shot Classifier: Another complex approach involves using a Multi-Layer Perceptron (MLP) instead of just an auto-encoder. The generator and discriminator function similarly to the approach described in section II-C2. However, the zero-shot classifier employs an

MLP classifier. It utilizes the MLP to generate an embedding vector of the same size as the embedding vector for Gene Ontology (GO) classes [8].

4) Language Models with a Zero-shot Classifier: Another approach to incorporate ZSL with GANs involves utilizing Large Language Models (LLMs) in combination with a zero-shot classifier.

In this method, a pre-trained LLM, such as BERT or GPT, is employed to encode and generate textual representations of the input data [32], [33]. The generator in the GAN framework generates synthetic data based on the learned representations from the LLM. However, the LLM is not directly involved in the generator's training process, and only in the zero-shot classifier.

The separate zero-shot classifier, which is based on the LLM, is trained to perform zero-shot classification of unseen classes. It takes the generated synthetic data as input and leverages the semantic and contextual information captured by the LLM to accurately classify the data into the appropriate unseen classes.

By utilizing the LLM-based zero-shot classifier as a separate component, this approach enables the generation and classification of synthetic data for unseen classes without directly involving the LLM in the training of the generator and discriminator. It leverages the shared knowledge and attributes of seen and unseen classes encoded in the LLM, facilitating accurate classification of the generated data [34], [35].

The integration of LLMs with a separate zero-shot classifier in the GAN framework presents a promising avenue for addressing the challenges posed by unseen classes and other data scenarios with limited samples. It harnesses the power of LLMs to generate and classify data, enabling the exploration and analysis of unseen classes in a ZSL setting [34], [36].

III. Methodology

This section presents the methodology for integrating ontology embedding and the ZSL classifier into the DP-CGANs framework. The objective is to enhance the performance of the GAN in generating and classifying data for both seen and unseen classes.

The overall structure of Onto(ZSL)-CGANs is illustrated in Figure 1, where the core architecture follows the DP-CGANs structure explained in [3]. We extend this architecture by incorporating ontology embeddings and a ZSL classifier.

During the sampling process, ontology embeddings serve as conditional vectors provided to the GAN. By incorporating these conditional vectors, the generator is guided to produce samples that align with the characteristics of specific classes defined in the ontology.

In the training process, the ZSL classifier takes the generated data samples as input and predicts whether they belong to seen or unseen classes. This additional component in the architecture enables the discrimination between the seen and unseen data. The predictions from the ZSL classifier are then fed back into the system and become part of the test data for the next iteration, similar to the discriminator in the GAN framework.

The extension of the DP-CGANs architecture with these two core improvements is expected to improve the quality of generated samples due to the utilization of both ontology embedding and the ZSL classifier.

A. Ontology Embedding

To incorporate semantic knowledge from ontologies into the GAN, we utilize ontology embeddings generated using owl2vec* for graph projection, DeepWalk for random walks, and word2vec for "word" embedding techniques through the mOWL library. These methods capture the semantic relationships and contextual information among ontology concepts and represent them as distributed vector representations. The embeddings are then used as conditional vectors for the generator and discriminator in the GAN.

The structure of the ontology embedding process is shown in Algorithm 1.

Algorithm 1 Implementation of Ontology Embedding

Require: ontology
Require: projection_type
Require: walking_type
Ensure: ontology is an .owl file.
  if projection_type is Owl2Vec* **then**
    $graph \leftarrow Owl2VecStar(ontology)$
  else if projection_type is DL2Vec **then**
    $graph \leftarrow DL2Vec(ontology)$
  end if
  if walking_type is DeepWalk **then**
    $walks \leftarrow DeepWalk(graph)$
  else if walking_type Node2Vec **then**
    $walks \leftarrow Node2Vec(graph)$
  end if
    return $Word2Vec(walks)$

B. Zero-Shot Learning Classification

We integrate a ZSL classifier component into the GAN, similar to the discriminator, to classify the synthetic data generated by the generator as either seen or unseen classes. The ZSL classifier component employs a pre-trained BERT-based LLM to leverage semantic and contextual information encoded in the LLM. This integration improves the generator's performance on unseen classes by utilizing shared knowledge between seen and unseen classes.

The similarity between the given results from the LLM is computed through cosine similarity as follows:

$$\hat{c} = \arg\max_{c \in C} cos(\phi(x), \phi(C))$$

Here, $\phi(x)$ represents the data to be predicted, while $\phi(C)$ represents the seen classes.

A discriminative threshold t is employed to enable the classification of seen classes while distinguishing them from

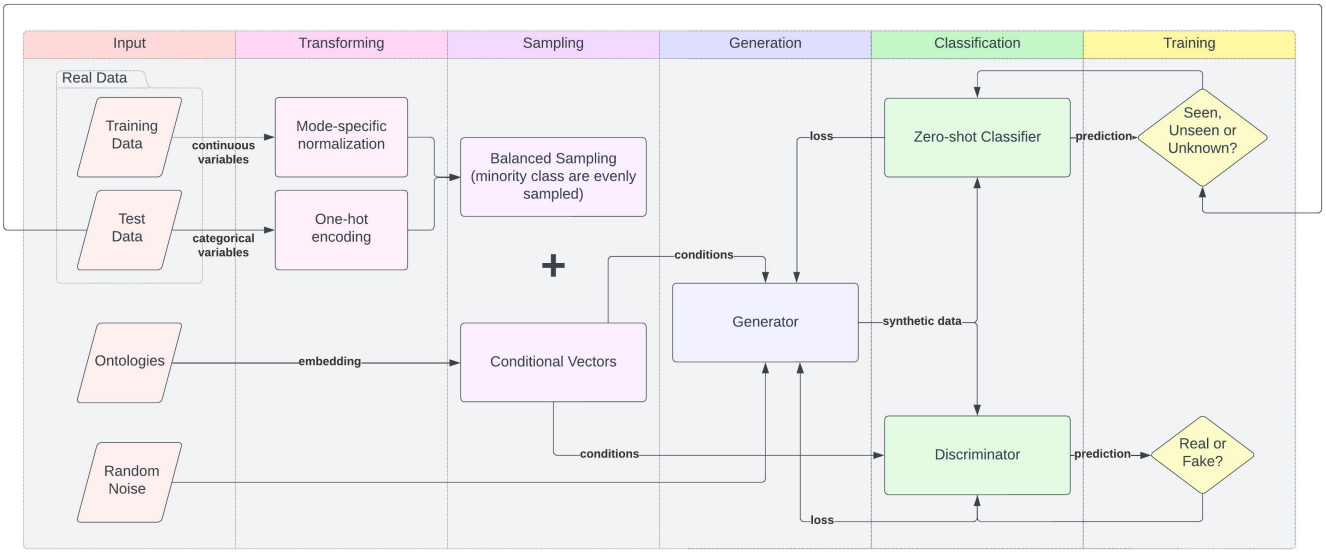


Fig. 1: The overall structure of OntoZSL-CGANs.

unseen classes based on the similarity $\hat{c}$. This threshold serves as a crucial mechanism to differentiate between the two categories and to facilitate the accurate classification of data points into their respective classes.

By setting an appropriate threshold value, we establish a clear boundary that allows for the identification and classification of seen classes while ensuring that instances falling below this threshold are correctly identified as belonging to unseen classes.

This approach ensures the effective separation and categorization of data points, thereby enabling the evaluation of both seen and unseen classes.

The classification process is defined as follows:

$$y(x) = \begin{cases} \hat{c} \geq t & \text{classify(x)} \\ else, & \text{Unseen} \end{cases}$$

Here, classify(x) returns the predicted label that has the highest similarity score.

The structure of the ZSL classification process is shown in Algorithm 2.

C. Generative Adversarial Network

In the GAN framework, ontology embeddings are incorporated as conditional vectors to the generator and discriminator during the sampling process. This allows the GAN to leverage these embeddings to generate synthetic data that aligns with the semantic relationships captured in the embeddings.

The synthetic data generated by the generator is then passed to the ZSL classifier component, similar to the discriminator.

This integrated approach combines the power of ontology embeddings, LLM-based ZSL classification, and DP-CGANs to generate and classify data for both seen and unseen classes, addressing the challenges posed by limited data scenarios.

Algorithm 2 Implementation of Zero-Shot Learning Classifier

Require: synthetic data D_S
Require: real data D_R
Require: threshold t
Ensure: $t > 0$
  $zsl \leftarrow$ Initialize $ZeroShotClassifier$
  $model \leftarrow$ Initialize $SentenceTransformer$ ▷ with BERT Model
  $E_R \leftarrow encode(model, D_R)$ ▷ $model$ encodes D_R
  $E_S \leftarrow encode(model, D_S)$ ▷ $model$ encodes D_S
  $sim \leftarrow sim(E_S, E_R)$ ▷ Compute Cosine Similarity between E_S and E_R
  if $sim['score'] \geq t$ **then** ▷ E_S and E_R are similar enough to be of the same seen class.
    $res \leftarrow sim['label']$ ▷ The seen Class
  else
    $res \leftarrow$ Unknown
  end if
    return res

IV. Experiments

This section describes the experiments conducted to evaluate the performance of the DP-CGANs framework in different configurations. We compare the baseline DP-CGANs framework, referred to as Baseline, with two variations:

- DP-CGANs with only ontology embeddings.
- DP-CGANs with both ontology embeddings and the zero-shot classifier.

The objective is to assess the impact of ontology embeddings and the zero-shot classification on the performance of the baseline, allowing us to explore the utility of ZSL in addressing the challenges posed by sparse data.

A. Experimental Setup

a) Baseline DP-CGANs Framework: To establish a baseline, we conducted experiments using the existing DP-CGANs framework in its original state, without incorporating any additional components or tweaking parameters. The baseline framework utilizes traditional supervised learning to train the generator and discriminator models on the available labeled data. We aim to evaluate the performance of this baseline in terms of its ability to generate realistic synthetic data and accurately classify seen classes.

b) Ontology-based DP-CGANs Framework: In this experiment, we enhance the DP-CGANs framework by integrating ontology embeddings into the GAN architecture, as explained in Section III-A.

We aim to evaluate how the addition of ontology embeddings improves the quality of synthetic data generation and classification performance for seen classes.

c) Ontology-based Zero-Shot Learning DP-CGANs: In this experiment, we further extend the Ontology-based DP-CGANs framework by integrating a zero-shot classifier component, as explained in Section III-B.

We aim to evaluate the effectiveness of the zero-shot classifier in accurately classifying both seen and unseen classes and its contribution to the overall performance of the DP-CGANs framework.

These experiments are conducted with the hyperparameters presented in Table IV.

B. Evaluation

As mentioned earlier, we conduct a performance comparison among three configurations: the baseline DP-CGANs framework, Ontology-based DP-CGANs, and Ontology-based ZSL DP-CGANs. Our evaluation encompasses multiple metrics, including the quality of the generated synthetic data, classification accuracy on seen classes, and zero-shot classification accuracy on unseen data.

a) Statistical Similarity: To assess the statistical similarity between the real and synthetic data, we employ well-established metrics. Given that the dataset used in our experiments contains binary data, excluding the labels, we focused our analysis on three specific tests: Kullback-Leibler (KL) Discrete Divergence [37], Pearson's Chi-square (CS) test [38], and the Point Biserial Correlation [39].

The Kullback-Leibler Discrete Divergence measures the dissimilarity between two probability mass functions (PMFs). This test evaluates the marginal distribution similarity between the two datasets by comparing the PMFs independently for each variable, regardless of the real and synthetic data. A KL divergence score approaching 1 indicates a higher level of similarity, suggesting that the synthetic data approximates the distribution of the real data more closely. The overall similarity of the data is approximated by averaging the similarity scores of all measured variables in the data [3].

Pearson's Chi-square test assesses the similarity of PMFs between real and synthetic data. It examines the null hypothesis that the PMFs of both datasets are identical, meaning that the frequencies of binary values in the synthetic data match those of the real data. A Chi-square score approaching 1 indicates a higher level of similarity, suggesting that the synthetic data captures the distribution characteristics of the real data more effectively [3].

The point biserial correlation measures the correlation between a binary variable and a continuous variable. It helps quantify the association between the binary data and the real/synthetic labels. A point biserial correlation value close to 0 indicates a strong positive correlation, meaning that the synthetic data successfully captures the relationship between the binary attributes and the labels.

Performing these statistical similarity tests allows us to quantitatively assess how closely the generated synthetic data aligns with the real data distribution. Higher values in the KL Divergence and CS tests and a lower point biserial correlation indicate a higher degree of similarity between the real and synthetic data, suggesting that the synthetic data accurately represents the underlying distribution and relationships in the real data.

b) Machine Learning Performance: To evaluate the quality of the generated synthetic data, we adopt a set of machine learning models, including Logistic Regression (LR), Decision Trees (DT), and Random Forests (RF). For this evaluation, we split the real dataset into a training set (75%) and a testing set (25%). The GAN is then trained on the training set, producing the synthetic data. Subsequently, we train the machine learning models on both the real and synthetic data separately and assess their performances on the real test data using evaluation metrics such as Area Under the Curve (AUC) and F1 scores. A smaller difference in the machine learning performance when using the synthetic data indicates higher quality and more realistic synthetic data compared to the real data.

This rigorous evaluation allows us to quantitatively compare the performances of different configurations and draw meaningful conclusions regarding the effectiveness of the three variations in generating realistic and accurate synthetic data for both common and uncommon data.

C. Dataset

The dataset used during the experiments is a phenotype dataset [40] specifically curated for rare diseases. It was generated by merging the Human Phenotype Ontology (HPO), Orphanet Rare Disease Ontology (ORDO), and Human Ontology of Medicine (HOOM) datasets (see Figure V). These datasets comprise a combination of real and synthetic phenotype data based on factual information extracted from the ontology data.

The HPO provides a comprehensive vocabulary of phenotype abnormalities and their relationships, ORDO offers structured information about rare diseases, and HOOM provides additional medical knowledge and terminology. By merging these datasets, we create a rich and diverse phenotype dataset that encompasses both common and rare phenotype attributes.

To generate synthetic phenotype data, we leverage the information present in the ontologies, ensuring that the synthetic data retains real-world phenotypes' semantic and contextual

Step	Model	Hyperparameter	Value
Transformation	Gaussian Mixture	Prior type for the weights' distribution	Dirichlet Process
		Max. num of Gaussian distribution	10
		Weight concentration prior	1×10^{-3}
		Weight threshold	1×10^{-3}
		Num of mixture components	≤ 10
Network training	-	Epochs	200
	-	Batch size	500
	Adam	Learning rate (α)	1×10^{-4}
	PacGAN	Pac	10
	WGAN-GP	Gradient penalty Factor (λ)	10
	Softmax	Non-negative scalar temperature (τ)	0.2
	LeakyReLU	Negative slope	0.2
	Dropout	Probability of an element is 0	0.5
Ontology Embedding	owl2vec*	Taxonomies	Yes
		Bidirectional Taxonomies	No
		Literals	No
		Relations	None
	DeepWalk	Num. Walks	15
		Walk Length	6
		Workers	4
		Learning Rate (α)	0.01
Zero-Shot Classifier	Bert Base Uncased	threshold	0.75

TABLE IV: The hyperparameters used for the DP-CGANS, Onto-CGANS and OntoZSL-CGANS experiments. The core structure of the hyperparameters are equivalent to those from the experiments of [3].

Dataset	#Classes	#Axioms	#SubClassesOf	Source	Access
HOOM	132.786	520.336	132.778	EBI	Public
ORDO	15.239	237.754	53.379	EBI	Public
HPO	20.958	192.334	21.408	EBI	Public

TABLE V: Descriptions of Ontology Datasets used to build the Phenotype dataset.

Dataset	#Instances	#Phenotypes	#Diseases	Diseases
Seen	12.000	30	12	Alpha-thalassemia-myelodysplastic syndrome Multiple myeloma Polycythemia vera Deafness-lymphedema-leukemia syndrome Acute panmyelosis with myelofibrosis Unclassified myelodysplastic syndrome Acute monoblastic/monocytic leukemia Acute myelomonocytic leukemia Acute erythroid leukemia Primary myelofibrosis Acute promyelocytic leukemia Adrenomyeloneuropathy
Unseen	60	30	3	Essential thrombocythemia Chronic myeloid leukemia Metachromatic leukodystrophy

TABLE VI: Description of the phenotype dataset.

characteristics. The synthetic data generation process involves inferring new phenotype instances based on existing ontological relationships and applying various data augmentation techniques. Although the phenotype dataset does not provide lab-based data, it provides a valuable resource for studying the effectiveness of ontology-based approaches and ZSL in the context of rare diseases.

The considered diseases for the experiments are from the leukemia family (see Table VI), where there is no overlap in classes between seen and unseen data.

V. Results

In this section, we present the results of the experiments conducted on the Phenotype dataset using three different models, as mentioned in Section IV. Each experiment was conducted twice, and the results were averaged to ensure reliability.

A. Statistical Similarity

To assess the statistical similarity between the real and synthetic data, we evaluated the performed tests, the results of which are presented in Table VII.

The KL discrete divergence and CS tests measure the similarity of each individual variable independently between the real and synthetic data. A higher score indicates a greater similarity between the synthetic data and the real data. The results indicate that both Onto-CGANs and OntoZSL-CGANs

slightly outperformed DP-CGANs in terms of KL discrete divergence and CS, although the differences were minimal.

On the other hand, the point-biserial correlation measures the difference of dependencies between a pair of variables. A lower score indicates that the synthetic data captures more of the dependencies present in the real data. The results show that all three methods demonstrated nearly equal performance in terms of correlation, with very small differences among them.

B. Machine Learning Performance

Figure 2 reports the evaluation results of the three machine learning models trained on synthetic data generated by the respective models and evaluated on the test set. To assess the model performance on imbalanced datasets, we placed emphasis on evaluating the AUC and F1 Score, as they are considered reliable metrics [41].

Comparing the synthetic data generated by DP-CGANs, Onto-CGANs, and OntoZSL-CGANs with the baseline, we observed that the models trained on DP-CGANs synthetic data showed improved performance for Onto-CGANs and OntoZSL-CGANs over the performance of DP-CGANs, although being more time-intensive due to the additional models that require training.

The F1 score for Onto-CGANs and OntoZSL-CGANs had an approximate improvement of 20% over DP-CGANs. Similarly, the AUC for Onto-CGANs and OntoZSL-CGANs had an approximate improvement of 15% over DP-CGANs.

Overall, it is evident that Onto-CGANs and OntoZSL-CGANs perform similarly to the real data, indicating that the generated data by Onto-CGANs and OntoZSL-CGANs should be similar to the real data.

VI. Discussion

A. Performance Comparison

In our study, we compared the performance of three different models: DP-CGANS, Onto-CGANS, and OntoZSL-CGANS. The results revealed that the synthetic data generated by Onto-CGANS and OntoZSL-CGANS outperformed DP-CGANS in terms of machine learning performance. The F1 score for Onto-CGANS and OntoZSL-CGANS was approximately 20% higher than that of DP-CGANS, indicating improved classification accuracy. Similarly, the AUC for Onto-CGANS and OntoZSL-CGANS was approximately 15% higher than that of DP-CGANS, indicating a better capture of the underlying data distribution. There was no significant difference in performance between Onto-CGANS and OntoZSL-CGANS, suggesting that the ZSL capability in OntoZSL-CGANS did not significantly impact its performance compared to Onto-CGANS.

B. Impact of Ontologies

The incorporation of ontological information in the generation process had a significant impact on the performance of the models. By leveraging ontologies, Onto-CGANS and OntoZSL-CGANS produced more realistic synthetic data that closely resembled the real data distribution. This improvement can be attributed to the structured and domain-specific knowledge encoded in the ontologies, which guided the generation process. The utilization of ontologies also contributed to the efficiency of Onto-CGANS, requiring fewer training epochs to achieve realistic results.

C. Impact of Zero-Shot Learning

The inclusion of ZSL into OntoZSL-CGANS aimed to enhance the model's ability to generate synthetic data for unseen classes. However, our results indicate that there was no significant difference in performance between Onto-CGANS and OntoZSL-CGANS. This suggests that the additional complexity introduced by ZSL did not provide a substantial advantage in terms of machine learning performance. Nonetheless, further exploration of ZSL techniques could potentially unlock additional benefits in generating synthetic data for previously unseen classes.

D. Limitations and Future Work

Despite the promising results, our study has certain limitations that warrant consideration. Firstly, the current implementation of our OntoZSL-CGANS model can only classify samples as "seen" or "unknown" classes and cluster the "seen" samples into their appropriate known classes. Future research should focus on utilizing ontologies to infer "unseen" classes based on their descriptions, enabling more comprehensive and diverse synthetic data generation without necessarily having seen the said data.

Secondly, the training and testing dataset used for the experiments is a synthetic dataset on leukemia generated using ontologies. Further experiments should be performed on real-world data to ensure better robustness and reliability.

Furthermore, our research highlights the need for more extensive investigations in the field of ZSL. As ZSL is still in its early stages of research, additional studies are necessary to establish reliable and robust techniques for effectively applying ZSL in the generation process.

VII. Conclusion

In this study, we investigated the effectiveness of integrating related ontologies into a GAN model for generating realistic personal health data. We compared the performance of three models: DP-CGANS, Onto-CGANS and OntoZSL-CGANS, and evaluated their machine learning performance using synthetic data generated from each model. Additionally, we explored the potential of ZSL for classifying generated synthetic personal health data on unseen classes.

Our research aimed to answer three primary research questions:

1) How can related ontologies be embedded into a GAN model to generate realistic personal health data?
2) How does an ontology-based GAN perform compared to a baseline GAN in generating personal health data?
3) How could zero-shot learning be used for the classification of generated synthetic personal health data using a GAN model on unseen classes?

Test	DP-CGANs	Onto-CGANs	OntoZSL-CGANs
KL Discrete Divergence	0.97485	0.98496	0.99002
Chi-Squared	0.95953	0.96765	0.96780
Point-Biserial Corr.	0.00047	0.00040	0.00041

TABLE VII: Results of measuring statistical similarities between real and synthetic data.

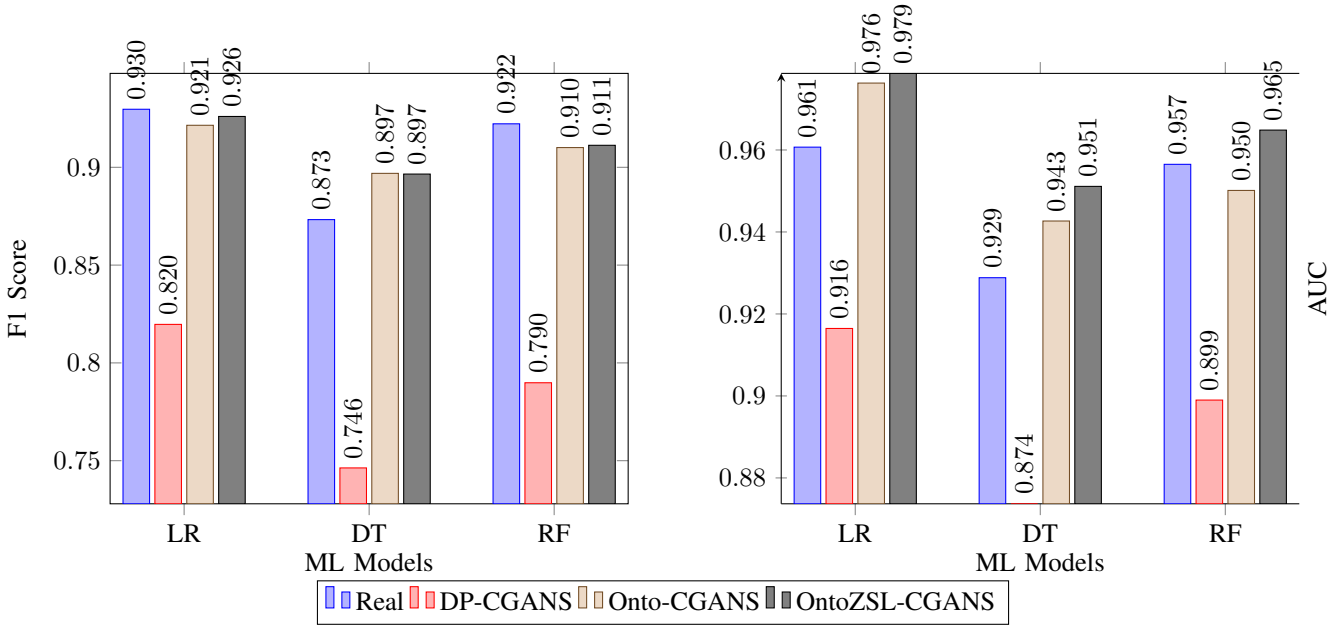


Fig. 2: Evaluation results of trained machine learning models on synthetic data using various generative models.

Through our experiments and analysis, we have obtained important insights regarding these research questions.

Firstly, the integration of related ontologies into the GAN model, as demonstrated by Onto-CGANS and OntoZSL-CGANS, proved to be an effective approach for generating realistic personal health data. By leveraging the structured and domain-specific knowledge encoded in the ontologies, the synthetic data generated by these models closely resembled the distribution of real data. This integration allowed the model to capture the complex relationships and dependencies present in personal health data, leading to improved performance in terms of classification accuracy and data distribution representation.

Secondly, the ontology-based GAN models (onto-CGANS and OntoZSL-CGANS) outperformed the baseline GAN model (DP-CGANS) in generating personal health data. The machine learning performance of the models trained on synthetic data generated by Onto-CGANS and OntoZSL-CGANS showed a significant improvement compared to DP-CGANS. The F1 score and AUC metrics indicated higher accuracy and a better capture of the underlying data distribution when using the synthetic data from the ontology-based models.

Lastly, our exploration of ZSL demonstrated its potential for the classification of generated synthetic personal health data on unseen classes. Although the ZSL capability implemented in OntoZSL-CGANS did not result in significant performance differences compared to Onto-CGANS, it provides a foundation for future advancements in generating synthetic data for previously unseen classes. Further research and development in ZSL techniques hold promise for enhancing the capabilities of GAN models in generating diverse and comprehensive synthetic personal health data.

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